Unit-3: STATISTICAL ANALYSIS FOR DATA SCIENCE

Descriptive Statistics:- Measures of Central Tendency: Mean, Median, Mode

Descriptive statistics is a type of statistical analysis that uses quantitative methods to summarize the features of a population sample. It is useful to present easy and exact summaries of the sample and observations using metrics such as mean, median, variance, graphs, and charts.

Measures of Central Tendency

The central tendency is defined as a statistical measure that may be used to describe a complete distribution or dataset with a single value, known as a measure of central tendency. Any of the central tendency measures accurately describes the whole data distribution. In the following sections, we will look at the central tendency measures, their formulae, applications, and kinds in depth.

- Mean
- Median
- Mode

Mean

Mean is the sum of all the components in a group or collection divided by the number of items in that group or collection. Mean of a data collection is typically represented as $\bar{x}$ (pronounced "x bar"). The formula for calculating the mean for ungrouped data to express it as the measure is given as follows:

For a series of observations:

$$\bar{x} = \Sigma x / n$$

where,

- $\bar{x}$ = Mean Value of Provided Dataset
- Σx = Sum of All Terms
- n = Number of Terms

Example: Weights of 7 girls in kg are 54, 32, 45, 61, 20, 66 and 50. Determine the mean weight for the provided collection of data.

Mean = $\Sigma x/n$

= (54 + 32 + 45 + 61 + 20 + 66 + 50)/7

= 328 / 7

= 46.85

Thus, the group's mean weight is 46.85 kg.

Median

Median of a data set is the value of the middle-most observation obtained after organizing the data in ascending order, which is one of the measures of central tendency. Median formula may be used to compute the median for many types of data, such as grouped and ungrouped data.

Ungrouped Data Median (n is odd): [(n +1)/2]th term

Ungrouped Data Median (n is even): [(n / 2)th term + ((n / 2) + 1)th term]/2

where,

- n = Number of Terms

Example: Weights of 7 girls in kg are 54, 32, 45, 61, 20, 66 and 50. Determine the median weight for the provided collection of data.

Arrange the provided data collection in ascending order: 20, 32, 45, 50, 54, 61, 66

Median = [(n + 1) / 2]th term

= [(7 + 1) / 2]th term

= 4th term

= 50

Thus, group's median weight is 50 kg.

Mode

Mode is one of the measures of central tendency, defined as the value that appears the most frequently in the provided data, i.e. the observation with the highest frequency is known as the mode of data. The mode formulae provided below can be used to compute the mode for ungrouped data.

Mode of Ungrouped Data: Most Repeated Observation in Dataset

Example: Weights of 7 girls in kg are 54, 32, 45, 61, 20, 45 and 50. Determine the mode weight for the provided collection of data.

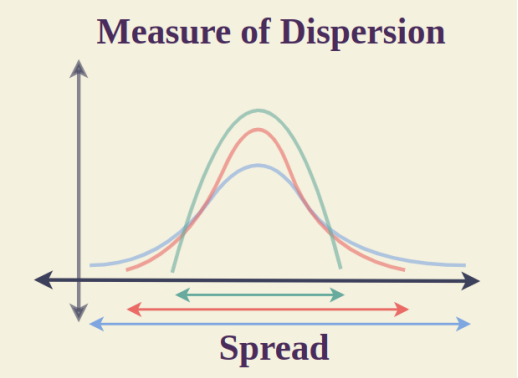
Mode = Most repeated observation in Dataset

= 45

Thus, group's mode weight is 45 kg.

Measures of Dispersion: Variance, Standard Deviation

Measures of Dispersion are used to represent the scattering of data. These are the numbers that show the various aspects of the data spread across various parameters.



Let’s learn about the measure of dispersion in statistics , its types, formulas, and examples in detail.

Dispersion in Statistics

Dispersion in statistics is a way to describe how spread out or scattered the data is around an average value. It helps to understand if the data points are close together or far apart.

Dispersion shows the variability or consistency in a set of data. There are different measures of dispersion like range, variance, and standard deviation.

Measure of Dispersion in Statistics

Measures of Dispersion measure the scattering of the data. It tells us how the values are distributed in the data set. In statistics, we define the measure of dispersion as various parameters that are used to define the various attributes of the data.

Variance: It is defined as the average of the square deviation from the mean of the given data set.

OR

Variance is a statistical measure that quantifies the spread or dispersion of a set of data points. It indicates how much the individual data points in a dataset differ from the mean (average) of the dataset

Types of Variance

We can define the variance of the given data in two types,

- Population Variance
- Sample Variance

Population Variance

Population variance is used to find the spread of the given population. The population is defined as a group of people and all the people in that group are part of the population. It tells us about how the population of a group varies with respect to the mean population.

When we want to find how each data point in a given population varies or is spread out then we use the population variance. It is used to give the squared distance of each data point from the population mean.

Population Variance Formula

The formula for population variance is written as,

$$\sigma^2 = \sum (x_i - \bar{x})^2/n$$

where,

- $\bar{x}$ is the mean of population data set
- **n** is the total number of observations

Population variance is mainly used when the entire population's data is available for analysis.

Sample Variance

If the population data is very large it becomes difficult to calculate the population variance of the data set. In that case, we take a sample of data from the given data set and find the variance of that data set which is called sample variance. While calculating the sample mean we make sure to calculate the sample mean, i.e. the mean of the sample data set not the population mean. We can define the sample variance as the mean of the square of the difference between the sample data point and the sample mean.

Sample Variance Formula

The formula of Sample variance is given by,

$$\sigma^2 = \sum (x_i - \bar{x})^2/(n - 1)$$

where,

- $\bar{x}$ is the mean of sample data set
- **n** is the total number of observations

Sample variance is typically used when working with data from a sample to infer properties about

Variance Solved Example

We can understand the concept of variance with the help of the example discussed below.

Find the population variance of the data {4,6,8,10}

Solution:

Mean = (4+6+8+10)/4 = 7

4	$(4-7)^2$	9
6	$(6-7)^2$	1
8	$(8-7)^2$	1
10	$(10-7)^2$	9

Variance = (9+1+1+9)/4 = 20/4 = 5

Thus, the variance of the data is 5

Standard Deviation: It is the square root of the arithmetic average of the square of the deviations measured from the mean.

OR

Standard deviation is a measure used in statistics to understand how the data points in a set are spread out from the mean value. It indicates the extent of the data's variation and shows how far individual data points deviate from the average.

Formula for Calculating Standard Deviation

Formula used for calculating Standard Deviation is discussed in the image below,

Population	Sample
$\sigma = \sqrt{\frac{\Sigma(x_i-\mu)^2}{n}}$	$S = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n-1}}$
μ - Population Average xi - Individual Population Value n - Total Number of Population	x - Sample Average x_i - Individual Population Value n - Total Number of Sample

Example: Find Standard Deviation of data set. X = {2, 3, 4, 5, 6}

Given,

- n = 5
- xi = {2, 3, 4, 5, 6}

We know,

Mean(μ) = (Sum of Observations)/(Number of Observations)

⇒ μ = (2 + 3 + 4 + 5 + 6)/ 5

⇒ μ = 4

using standard deviation formula

$$\sigma^2 = \sum_i^n (x_i - \bar{x})^2/n$$

$$\Rightarrow \sigma^2 = 1/n[(2-4)^2 + (3-4)^2 + (4-4)^2 + (5-4)^2 + (6-4)^2]$$

$$\Rightarrow \sigma^2 = 10/5 = 2$$

$$\text{Thus, } \sigma = \sqrt{(2)} = 1.414$$

Understanding Data Distribution: Normal Distribution, Skewness, Kurtosis

Normal Distribution is the most common or normal form of distribution of Random Variables, hence the name "normal distribution." It is also called **Gaussian Distribution** in Statistics or Probability. We use this distribution to represent a large number of random variables. It serves as a foundation for statistics and probability theory.

Probability Density Formula for Normal Distribution

$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

where,

- x is **Random Variable**
- μ is **Mean**
- σ is **Standard Deviation**

Example 1: Find the probability density function of the normal distribution of the following data. x = 2, μ = 3 and σ = 4.

Step 1: Calculate $\sigma\sqrt{2\pi}$

First, calculate $\sigma\sqrt{2\pi}$:

$$\sigma = 4$$
$$\sqrt{2\pi} \approx 2.5066$$

Now multiply:

$$\sigma\sqrt{2\pi} = 4 \times 2.5066 \approx 10.0264$$

Step 2: Calculate $(x-\mu)^2$

Now calculate $(x-\mu)^2$:

$$x - \mu = 2 - 3 = -1$$
$$(x-\mu)^2 = (-1)^2 = 1$$

Step 3: Calculate $-\frac{(x-\mu)^2}{2\sigma^2}$

Now compute $-\frac{(x-\mu)^2}{2\sigma^2}$:

$$\sigma^2 = 4^2 = 16$$
$$-\frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2 \times 16} = -\frac{1}{32}$$

Step 4: Calculate $e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Now calculate $e^{-\frac{(x-\mu)^2}{2\sigma^2}}$:

$$e^{-\frac{1}{32}} \approx e^{-0.03125} \approx 0.969$$

Step 5: Combine all parts to find $f(x)$

Now plug everything into the PDF formula:

$$f(2) = \frac{1}{10.0264} \times 0.969$$

Calculating:

$$f(2) \approx 0.0967$$

Skewness

Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. In simpler terms, it indicates whether the data is concentrated more on one side of the mean compared to the other side.

Why is skewness important?

Understanding the skewness of data is crucial for several reasons:

1. **Modeling Assumptions**: As you mentioned, linear models often assume that the independent and target variables have similar distributions. Skewness information helps in assessing the degree of deviation from this assumption, allowing for appropriate adjustments or transformations to be made to improve model performance.
2. **Feature Engineering**: Skewed data can significantly impact the performance of machine learning models. Knowing the skewness helps in identifying which features may need transformation (e.g., log transformation) to make the data more suitable for modeling.
3. **Prediction Accuracy**: Skewness provides insights into the distribution of data values, which can influence how well a model generalizes to new, unseen data. By understanding the skewness, one can anticipate how the model might perform on different segments of the data and make adjustments accordingly.
4. **Outlier Detection**: Skewed distributions often indicate the presence of outliers, which can have a significant impact on model estimation and inference. Knowing the direction of skewness helps in identifying where outliers are more likely to occur, guiding the outlier detection and treatment process.
5. **Interpretation of Results**: Skewness information aids in the interpretation of model results. For instance, in your example of predicting mpg based on car horsepower, knowing that the data is positively skewed suggests that the model may perform better for cars with lower horsepower, which informs decision-making and further analysis.

Coefficient of Skewness

- Pearson's first coefficient of skewness involves subtracting the mode from the mean and then dividing the difference by the standard deviation. This method is effective for data with a prominent mode but may not be suitable for datasets with low mode or multiple modes. In such cases, Pearson's second coefficient may be preferable.
  First coefficient : Mean–Mode/StandardDeviation
- Pearson's second coefficient of skewness is calculated by subtracting the median from the mean, multiplying the difference by 3, and dividing the product by the standard deviation.
  Second coefficient : 3(Mean–Median)/StandardDeviation Otherwise, we can use this formula too,

Skewness Formula = $\Sigma(X–\bar{X})$

Example: Calculating Pearson's median skewness

Pearson's median skewness of the number of sunspots observed per year:

- Mean = 48.6
- Median = 39
- Standard deviation = 39.5

Calculation

$$\text{Pearson's median skewness} = 3 \times \frac{(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

$$\text{Pearson's median skewness} = 3 \times \frac{(48.6 - 39)}{39.5}$$

$$\text{Pearson's median skewness} = 0.73$$

Interpretation of skewness

Skewness is a measure of the asymmetry of a distribution. It tells us about the extent to which the data deviates from a symmetric distribution. Here's how to interpret skewness:

1. **Skewness value around 0**: **A skewness value close to zero (between -0.5 and 0.5)** indicates that the distribution is approximately symmetric. This means that the data are evenly distributed around the mean, with roughly equal frequencies of values on both sides.
2. **Negative skewness**: If the **skewness is negative (less than -0.5)**, it indicates that the left tail of the distribution is longer or stretched out compared to the right tail. In other words, the majority of the data points are concentrated on the right side of the distribution, with a few extremely low values dragging the mean to the left.
3. **Positive skewness**: Conversely, if the **skewness is positive (greater than 0.5)**, it suggests that the right tail of the distribution is longer or stretched out relative to the left tail. In this case, most of the data points are clustered on the left side of the distribution, with a few extremely high values pulling the mean to the right.
4. **Magnitude of skewness**: The magnitude of the skewness value indicates the degree of asymmetry. Larger positive or negative values imply greater asymmetry. **Skewness values less than -1 or greater than 1 are considered highly skewed.**

Type of Skewness

Various types of skewness used in mathematics are,

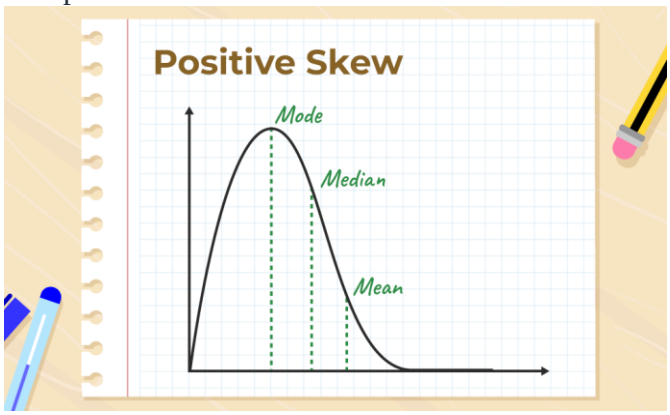
- Positive Skewness
- Negative Skewness
- Zero Skewness

Positive Skewness

Positive Skewness means the tail on the right side of the distribution is longer. The mean and median will be greater than the mode.

Condition for positive skewness = **Mean > Median >Mode**

The positive curve of skewness is shown in the image below,



Let's take an example of the income distribution where a few people earn very high incomes and the majority earn lower incomes. so, this is often positively skewed. Analyzing skewed data can provide valuable insights into the underlying causes and potential solutions or interventions.

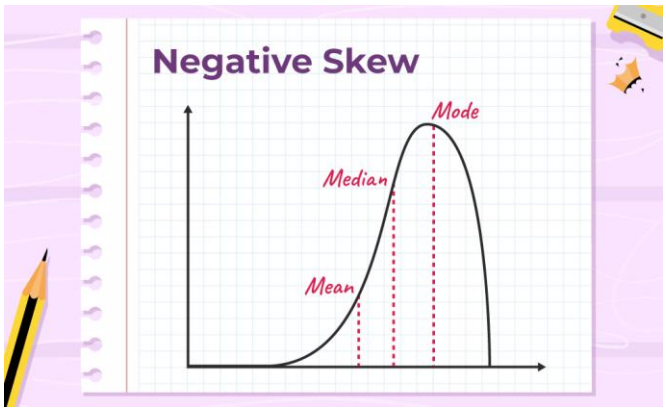
Negative Skewness

Negative Skewness means when the tail of the left side of the distribution is longer than the tail on the right side. The mean and median will be less than the mode.

Condition for negative skewness is,

Mode > Median > Mean

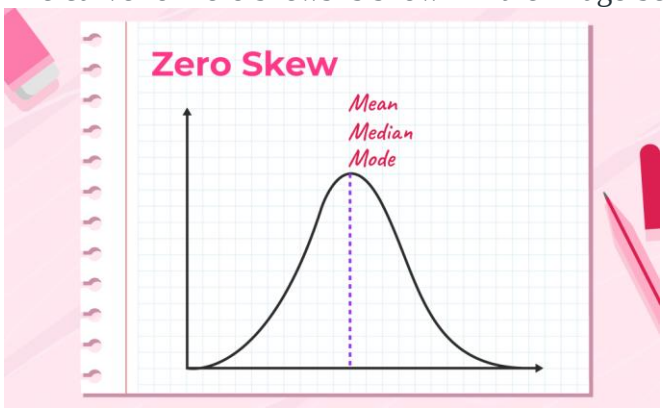
The curve shows negative skewness in the image below,

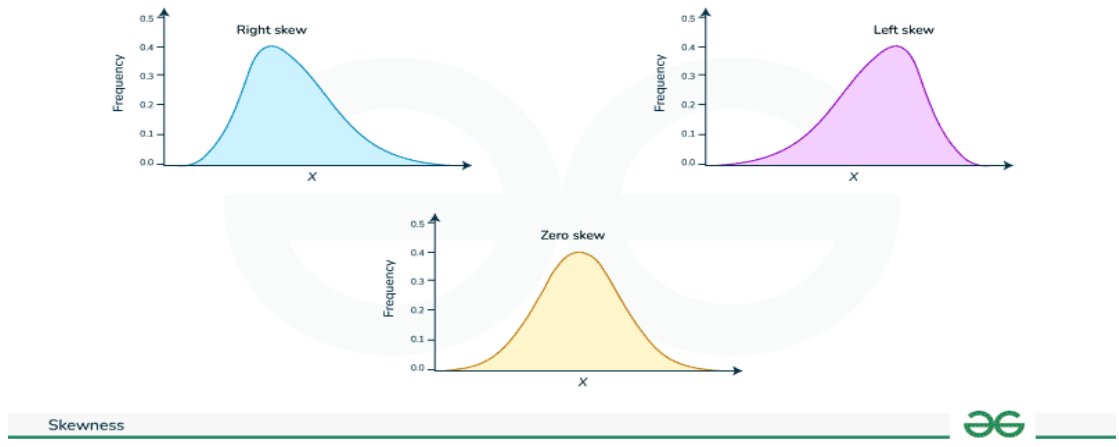


Let’s take an example of a match, during the match most of the players of a particular team scored runs above 50 and only a few of them scored below 10. In such a case, the data is generally represented with the help of a negatively skewed distribution. And this data is helpful to analyze the game’s performance.

Zero Skewness

It is also known as a “symmetric distribution”. It signifies that distribution of data is evenly distributed around the mean, with no long tails on either end of the distribution
Condition for zero skewness is **Mean = Mode = Median**
The curve for zero skews is shown in the image below,





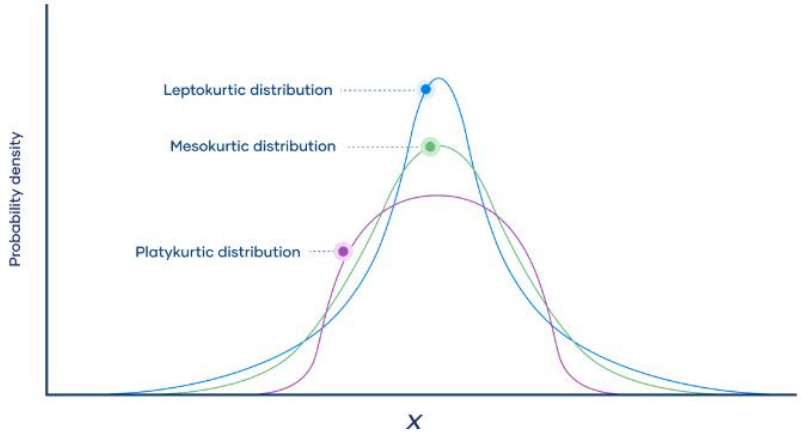
Kurtosis

In data science, kurtosis is a statistical measure used to describe the shape of a distribution's tails in comparison to a normal distribution. Specifically, kurtosis helps determine if the data have heavy tails (more outliers) or light tails (fewer outliers) relative to a normal distribution.

Types of Kurtosis
Mesokurtic (Kurtosis ≈ 3): This is the kurtosis of a normal distribution. It serves as a reference point, meaning that the data have moderate tail weight and outliers are as expected.

Leptokurtic (Kurtosis > 3): A distribution with positive kurtosis has heavy tails and a sharp peak, meaning more data are located in the tails (higher frequency of extreme values). This often implies more outliers.

Platykurtic (Kurtosis < 3): A distribution with negative kurtosis has light tails and a flatter peak, meaning fewer extreme values than expected under a normal distribution.



Formula

The formula for sample kurtosis K is generally:

$$K = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{X_i - \bar{X}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

where:

- X_i is each individual value,
- $\bar{X}$ is the sample mean,
- s is the sample standard deviation,
- n is the sample size.

Example: Calculating Kurtosis of a Dataset

Suppose we have a small dataset:

X={2,4,4,4,5,5,7,9}

We'll go through each step to calculate the kurtosis of this data.

Step 1: Calculate the Mean of the Dataset

The mean ($\bar{X}$) is calculated as:

$$\bar{X} = \frac{\sum X}{n} = \frac{2+4+4+4+5+5+7+9}{8} = \frac{40}{8} = 5$$

Step 2: Calculate the Standard Deviation

The standard deviation s is calculated by finding the squared differences from the mean, summing them, dividing by $n-1$ for a sample, and taking the square root:

$$s = \sqrt{\frac{(2-5)^2+(4-5)^2+(4-5)^2+(4-5)^2+(5-5)^2+(5-5)^2+(7-5)^2+(9-5)^2}{8-1}}$$

$$= \sqrt{\frac{9+1+1+1+0+0+4+16}{7}} = \sqrt{\frac{32}{7}} \approx 2.14$$

Step 3: Calculate the Fourth Moment (Kurtosis Formula)

Now, we use the following formula for kurtosis K:

$$K = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{X_i - \bar{X}}{s}\right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

Where:

- X_i are the data points,
- $\bar{X}$ is the mean,
- s is the standard deviation, and
- $n = 8$ in this case.

Let's calculate $\left(\frac{X_i - \bar{X}}{s}\right)^4$ for each X_i:

For each data point in X:

1. $(2-5)^4/s^4 = (-3/2.14)^4 \approx 2.73$
2. $(4-5)^4/s^4 = (-1/2.14)^4 \approx 0.04$
3. (Repeated for each point...)

Inferential Statistics:-Hypothesis Testing: Null and Alternative Hypotheses

Inferential statistics can be defined as a field of statistics that uses analytical tools for drawing conclusions about a population by examining random samples. The goal of inferential statistics is to make generalizations about a population. In inferential statistics, a statistic is taken from the sample data (e.g., the sample mean) that used to make inferences about the population parameter (e.g., the population mean).

What Is Hypothesis Testing in Statistics?

Hypothesis testing is a structured method used to determine if the findings of a study provide evidence to support a specific theory relevant to a larger population.

Hypothesis Testing is a type of statistical analysis in which you put your assumptions about a population parameter to the test. It is used to estimate the relationship between 2 statistical variables.

Let's discuss few examples of statistical hypothesis from real-life -

- A teacher assumes that 60% of his college's students come from lower-middle-class families.
- A doctor believes that 3D (Diet, Dose, and Discipline) is 90% effective for diabetic patients.

Statistical analysts validate assumptions by collecting and evaluating a representative sample from the data set under study.

The process of hypothesis testing involves four key steps: defining the hypotheses, developing a plan for analysis, examining the sample data, and interpreting the final results.

Importance of Hypothesis Testing in Data Analysis

Here is what makes hypothesis testing so important in data analysis and why it is key to making better decisions:

- Avoiding Misleading Conclusions (Type I and Type II Errors)

One of the biggest benefits of hypothesis testing is that it helps you avoid jumping to the wrong conclusions. For instance, a Type I error could occur if a company launches a new product thinking it will be a hit, only to find out later that the data misled them. A Type II error might happen when a company overlooks a potentially successful product because their testing wasn’t thorough enough. By setting up the right significance level and carefully calculating the p-value, hypothesis testing minimizes the chances of these errors, leading to more accurate results.

- Making Smarter Choices

Hypothesis testing is key to making smarter, evidence-based decisions. Let's say a city planner wants to determine if building a new park will increase community engagement. By testing the hypothesis using data from similar projects, they can make an informed choice. Similarly, a teacher might use hypothesis testing to see if a new teaching method actually improves student performance. It's about taking the guesswork out of decisions and relying on solid evidence instead.

- Optimizing Business Tactics

In business, hypothesis testing is invaluable for testing new ideas and strategies before fully committing to them. For example, an e-commerce company might want to test whether offering free shipping increases sales. By using hypothesis testing, they can compare sales data from customers who received free shipping offers and those who didn't. This allows them to base their business decisions on data, not hunches, reducing the risk of costly mistakes.

Hypothesis Testing Formula

$Z = (\bar{x} – \mu0) / (\sigma / \sqrt{n})$

- Here, $\bar{x}$ is the sample mean,
- $\mu0$ is the population mean,
- σ is the standard deviation,
- n is the sample size.

How Hypothesis Testing Works?

An analyst performs hypothesis testing on a statistical sample to present evidence of the plausibility of the null hypothesis. Measurements and analyses are conducted on a random sample of the population to test a theory. Analysts use a random population sample to test two hypotheses: the null and alternative hypotheses.

The null hypothesis is typically an equality hypothesis between population parameters; for example, a null hypothesis may claim that the population means return equals zero. The alternate hypothesis is essentially the inverse of the null hypothesis (e.g., the population means the return is not equal to zero). As a result, they are mutually exclusive, and only one can be correct. One of the two possibilities, however, will always be correct.

Null Hypothesis and Alternative Hypothesis

The Null Hypothesis is the assumption that the event will not occur. A null hypothesis has no bearing on the study's outcome unless it is rejected.

H0 is the symbol for it, and it is pronounced H-naught.

The Alternate Hypothesis is the logical opposite of the null hypothesis. The acceptance of the alternative hypothesis follows the rejection of the null hypothesis. H1 is the symbol for it.

Let's understand this with an example.

A sanitizer manufacturer claims that its product kills 95 percent of germs on average.

To put this company's claim to the test, create a null and alternate hypothesis.

H0 (Null Hypothesis): Average = 95%.

Alternative Hypothesis (H1): The average is less than 95%.

Another straightforward example to understand this concept is determining whether or not a coin is fair and balanced. The null hypothesis states that the probability of a show of heads is equal to the likelihood of a show of tails. In contrast, the alternate theory states that the probability of a show of heads and tails would be very different.

Hypothesis Testing Calculation With Examples

Let's consider a hypothesis test for the average height of women in the United States. Suppose our null hypothesis is that the average height is 5'4". We gather a sample of 100 women and determine their average height is 5'5". The standard deviation of population is 2.

To calculate the z-score, we would use the following formula:

$z = (\bar{x} – \mu 0) / (\sigma / \sqrt{n})$

$z = (5'5" - 5'4") / (2" / \sqrt{100})$

$z = 0.5 / (0.045)$

$z = 11.11$

We will reject the null hypothesis as the z-score of 11.11 is very large and conclude that there is evidence to suggest that the average height of women in the US is greater than 5'4".

p-values t-tests: One-Sample, Independent, Paired t-tests

What is a T-Test?

The T-test is a hypothesis testing method that helps in testing the significance of two or more groups and determining the important differences between the groups being compared. It is a variation of inferential statistics and is mostly used with datasets that possess a normal distribution but with unidentified variances. In terms of hypothesis testing, the t-value is a type of test statistic that is derived from the data sample, allowing to compare the sample with a null hypothesis i.e. a hypothesis with no strong distinction between selected populations whatsoever.

There are several approaches to t-testing depending on what data is being processed and the kind of analysis that is required, and if there are any specific data values required to accurately calculate the t-test. if there is a significant difference between the means of two groups, which may be related to certain features. It is mostly used when the dataset follows a normal distribution. It allows testing of an assumption applicable to a population, comparing the average values of the two data sets and determining if they came from the same population. A t-test can also be best described as a mathematical method of establishing a problem statement by making an assumption that the null hypothesis is for two equal means based on samples taken from each of the two datasets.

Assumptions for Performing T-Test

The following are the assumptions to be taken care of while performing a T-Test :

- The data being used for the test should follow a continuous or ordinal scale
- The data being used should be a random sample, that is the data is collected from a representative, randomly selected portion of the total population in the dataset.
- The data, when plotted, should result in a normal distribution or a bell-shaped distribution curve.
- Variance only exists when the standard deviations of samples are approximately equal.

Which T-Test To Use and When

- **Paired T-Test**: It tests whether the mean of the differences between the dependent or paired observations is equivalent to a target value. If the groups come from a single population, measured before and after the experimental treatment, a paired t-test should be performed.
- **Two-Sample T-Test:** It tests whether the difference between the means of two independent populations is equal to the target value. If the groups come from two different populations i.e two different species or people from two separate cities, then a two-sample t-test shall be performed.
- **One-Sample T-Test:** It tests whether the mean of a single population is equal to its target value. If there is just one group being compared against a standard value, in such a case a one-sample t-test must be performed.

What is a p-Value?

A p-value stands for the measure of the probability that an observed difference could have or has occurred by random chance. The lower the p-value, the greater the statistical significance of the observed difference. A P-value can be used as an alternative to the pre-selected confidence levels for hypothesis testing. The p-value is calculated using an integral calculus from the area under the probability distribution curve for all values that are as far from the reference value as the observed value is and relative to the total area under the probability distribution curve. The greater the difference between two observed values, the less likely it is that the difference might be due to a simple random chance, as reflected by a lower p-value. The significance level here is stated in advance to determine how small the p-value has to be in order to reject the null hypothesis. As everyone uses a different level of significance when examining a question, one might at times face difficulty comparing the results from two different tests. P-values provide a solution to such a problem.

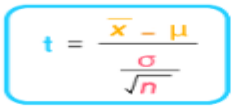
One-Sample T-Test Formula

For comparing the mean of a population $\bar{x}$ from n samples, with a specified theoretical mean μ, we use a one-sample t-test.

$$t = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where σ/√n is the standard error

t-Test Formula



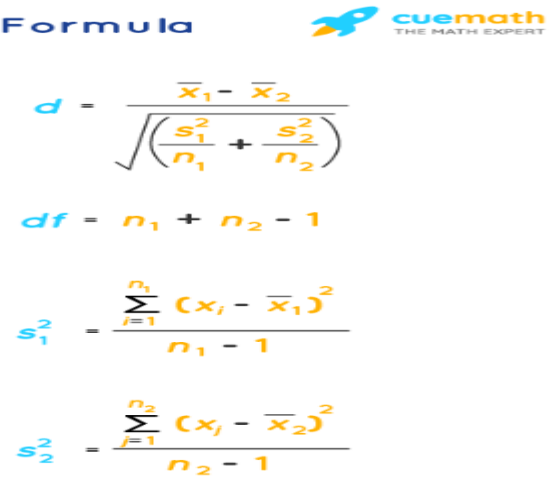
where,
'x' bar is the mean of the sample,
μ is the assumed mean,
σ is the standard deviation
and n is the number of observations

Independent Sample T-Test

Students t-test is used to compare the mean of two groups of samples. It helps evaluate if the means of the two sets of data are statistically significantly different from each other.

$$t = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})}}$$

T-test Formula



where

- t = Student's t-test
- x_1 = mean of first group
- x_2= mean of second group
- s_1 = standard deviation of group 1
- s_2 = standard deviation of group 1
- n_1= number of observations in group 1
- n_2= number of observations in group 2

Paired Samples T-Test

Whenever two distributions of the variables are highly correlated, they could be pre and post test results from the same people. In such cases, we use the paired samples t-test.

$$t = \frac{\Sigma(x_1 - x_2)}{\frac{s}{\sqrt{n}}}$$

where

t = Student's t-test

$x_1 - x_2$ = Difference mean of the pairs

s= standard deviation

n = sample size

Examples Using t-test Formula

Example 1: Calculate a t-test for the following data of the number of times people prefer coffee or tea in five time intervals.

Coffee	Tea
4	3
5	8
7	6
6	4
9	7

Solution: let x_1 be the sample of data that prefers coffee and x_2 be the sample of data that prefers tea.

let us find the mean, variance and the SD

x_1	$(x_1 - \overline{x_1})$	$(x_1 - \overline{x_1})^2$	x_2	$(x_2 - \overline{x_2})$	$(x_2 - \overline{x_2})^2$
4	-2.2	4.84	3	-2.6	6.76
5	-1.2	1.44	8	2.4	5.76
7	0.8	0.64	6	0.4	0.16
6	-0.2	0.04	4	-1.6	2.56
9	2.8	7.84	7	1.4	1.96
6.2		**14.8**	**5.6**		**17.20**

$\overline{x_1} = 31/5 = 6.2$

$\overline{x_2} = 28/5 = 5.6$

$\Sigma(x_1-\overline{x_1})^2 = 14.8$

$\Sigma(x_2-\overline{x_2})^2 = 17.2$

$S_1 = 14.8/4 = 3.7$

$S_2 = 17.2/4 = 4.3$

According to the t-test formula,

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})}}$$

Applying the known values in the t-test formula, we get

$$t = \frac{6.2 - 5.6}{\sqrt{(\frac{3.7}{5} + \frac{4.3}{5})}}$$

$$= \frac{0.6}{\sqrt{1.6}} = 0.6/1.26 = 0.47$$

t = 0.47

Example 2: A company wants to improve its sales. The previous sales data indicated that the average sale of 25 salesmen was $50 per transaction. After training, the recent data showed an average sale of $80 per transaction. If the standard deviation is $15, find the t-score. Has the training provided improved the sales?

Solution:

H_0accepted hypothesis:the population mean = the claimed value⇒ $\mu = \mu_O$

H_0alternate hypothesis: the population mean not equal to the claimed value⇒ $\mu \neq \mu_O$

t-test formula for independent test is $t = \frac{m - \mu}{\frac{s}{\sqrt{n}}}$

Mean sale = 80, μ = 50, s= 15 and n= 25

substituting the values, we get t= (80-50)/(15/√25)

t = (30 ×5)/10 = 10

looking at the t-table we find 10 > 1.711 . (I.e. CV for α = 0.05). ∴ the accepted hypothesis is not true. Thus we conclude that the training boosted the sales.

Example 3: A pre-test and post-test conducted during a survey to find the study hours of Patrick on weekends. Calculate the t-score and determine (for α = 0.25) if the pre-test and post-test surveys are significantly different?

Pre-test(X)	Post-test(Y)	X-Y	$(X-Y)^2$
1	2	-1	1
2	4	-2	4

Solution:

According to the t-test formula, we know that $t = \frac{\Sigma X - Y}{\frac{s}{\sqrt{n}}}$

Σ(X-Y)= -3 = 3

s= $\Sigma(X-Y)^2/(n-1) = 5^2/1 = 25$

t= 3/(25/2) = 6/25 = 0.24

here degree of freedom is n-1 = 2-1 =1 and the corresponidng critical value in the t-table for α= 0.25, is 1.

t < CV.

Therefore the scores are not significantly different.

P-value Formula

We Know that P-value is a statistical measure, that helps to determine whether the hypothesis is correct or not. P-value is a number that lies between 0 and 1. The level of significance(α) is a predefined threshold that should be set by the researcher. It is generally fixed as 0.05. The formula for the calculation for P-value is

Step 1: Find out the test static Z is

$$z = \frac{\hat{p} - p0}{\sqrt{\frac{p0(1-p0)}{n}}}$$

Where,

$\hat{p} = \text{Sample Proportion}$
P0 = assumed population proportion in the null hypothesis

N = sample size

Step 2: Look at the Z-table to find the corresponding level of P from the z value obtained.

P-value Table

The below-mentioned P-value table helps in determining the hypothesis according to the p-value.

P-value	Description	Hypothesis Interpretation
P-value ≤ 0.05	It indicates the null hypothesis is very unlikely.	Rejected
P-value > 0.05	It indicates the null hypothesis is very likely.	Accepted or it "fails to reject".
P-value > 0.05	The P-value is near the cut-off. It is considered as marginal	The hypothesis needs more attention.

P-Value Example

An example to find the P-value is given here.

Question: A statistician wants to test the hypothesis H_0: μ = 120 using the alternative hypothesis Hα: μ > 120 and assuming that α = 0.05. For that, he took the sample values as

n =40, σ = 32.17 and $\bar{x}$ = 105.37. Determine the conclusion for this hypothesis?

Solution:

We know that,

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Now substitute the given values

$$\sigma_{\bar{x}} = \frac{32.17}{\sqrt{40}} = 5.0865$$

Now, using the test static formula, we get

t = (105.37 − 120) / 5.0865

Therefore, t = −2.8762

Using the Z-Score table, we can find the value of P(t>−2.8762)

From the table, we get

P (t<−2.8762) = P(t>2.8762) = 0.003

Therefore,

If P(t>−2.8762) =1− 0.003 =0.997

P− value =0.997 > 0.05

Therefore, from the conclusion, if p>0.05, the null hypothesis is accepted or fails to reject.

Hence, the conclusion is "fails to reject H_0."

Example 1: A statistician is testing the hypothesis H0: μ = 120 using the approach of alternative hypothesis Hα: μ > 120 and assuming that α = 0.05. The sample values that he took are as n =40, σ = 32.17 and $\bar{x}$ = 105.37. What is the conclusion for this hypothesis?

Solution:

We know that,

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Now substitute the given values

$$\sigma_{\bar{x}} = \frac{32.17}{\sqrt{40}} = 5.0865$$

As per the test static formula, we get

t = (105.37 – 120) / 5.0865

Therefore, t = -2.8762

Using the Z-Score table, finding the value of P(t > -2.8762)

we get,

P (t < -2.8762) = P(t > 2.8762) = 0.003

Therefore,

If P(t > -2.8762) =1 - 0.003 =0.997

P- value =0.997 > 0.05

As the value of p > 0.05, the null hypothesis is accepted.

Therefore, the null hypothesis is accepted.

Example 2: P-value is 0.3105. If the level of significance is 5%, find if we can reject the null hypothesis.

Solution: Looking at the P-value table, the p-value of 0.3105 is greater than the level of significance of 0.05 (5%), we fail to reject the null hypothesis.

Analysis of Variance (ANOVA): One-way and Two-way ANOVA

What is analysis of variance (ANOVA)? Analysis of Variance (ANOVA) is a statistical formula used to compare variances across the means (or average) of different groups. A range of scenarios use it to determine if there is any difference between the means of different groups.

ANOVA Test Definition

ANOVA test can be defined as a type of test used in hypothesis testing to compare whether the means of two or more groups are equal or not. This test is used to check if the null hypothesis can be rejected or not depending upon the statistical significance exhibited by the parameters. The decision is made by comparing the ANOVA test statistic with the critical value.

ANOVA Formula:

ANOVA Test Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Value
Between Groups	$SSB = \Sigma n_j(\overline{X}_j - \overline{X})^2$	$df_1 = k - 1$	$MSB = SSB / (k - 1)$	$f = MSB / MSE$
Error	$SSE = \Sigma\Sigma(X - \overline{X}_j)^2$	$df_2 = N - k$	$MSE = SSE / (N - k)$	
Total	$SST = SSB + SSE$	$df_3 = N - 1$		

ANOVA Table

The ANOVA formulas can be arranged systematically in the form of a table. This ANOVA table can be summarized as follows:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Value
Between Groups	$SSB = \Sigma n_j(\overline{X}_j - \overline{X})^2$	$df_1 = k - 1$	$MSB = SSB / (k - 1)$	$f = MSB / MSE$
Error	$SSE = \Sigma\Sigma(X - \overline{X}_j)^2$	$df_2 = N - k$	$MSE = SSE / (N - k)$	
Total	$SST = SSB + SSE$	$df_3 = N - 1$		

One Way ANOVA

The one way ANOVA test is used to determine whether there is any difference between the means of three or more groups. A one way ANOVA will have only one independent variable. The hypothesis for a one way ANOVA test can be set up as follows:

Null Hypothesis, H_0**:** $\mu_1 = \mu_2 = \mu_3 = \ldots = \mu_k$

Alternative Hypothesis, H_1**:** The means are not equal

Decision Rule: If test statistic > critical value then reject the null hypothesis and conclude that the means of at least two groups are statistically significant.

The steps to perform the one way ANOVA test are given below:

- **Step 1:** Calculate the mean for each group.
- **Step 2:** Calculate the total mean. This is done by adding all the means and dividing it by the total number of means.
- **Step 3:** Calculate the SSB.
- **Step 4:** Calculate the between groups degrees of freedom.
- **Step 5:** Calculate the SSE.
- **Step 6:** Calculate the degrees of freedom of errors.
- **Step 7:** Determine the MSB and the MSE.
- **Step 8:** Find the f test statistic.
- **Step 9:** Using the f table for the specified level of significance, α, find the critical value. This is given by $F(\alpha, df_1. df_2)$.
- **Step 10:** If f > F then reject the null hypothesis.

Two Way ANOVA

The two way ANOVA has two independent variables. Thus, it can be thought of as an extension of a one way ANOVA where only one variable affects the dependent variable. A two way ANOVA test is used to check the main effect of each independent variable and to see if there is an interaction effect between them. To examine the main effect, each factor is considered separately as done in a one way ANOVA. Furthermore, to check the interaction effect, all factors are considered at the same time. There are certain assumptions made for a two way ANOVA test. These are given as follows:

- The samples drawn from the population must be independent.
- The population should be approximately normally distributed.
- The groups should have the same sample size.
- The population variances are equal

Suppose in the two way ANOVA example, as mentioned above, the income groups are low, middle, high. The gender groups are female, male, and transgender. Then there will be 9 treatment groups and the three hypotheses can be set up as follows:

H01: All income groups have equal mean anxiety.

H11: All income groups do not have equal mean anxiety.

H02: All gender groups have equal mean anxiety.

H12: All gender groups do not have equal mean anxiety.

H03: Interaction effect does not exist

H13: Interaction effect exists.

- **Example 1:** Three types of fertilizers are used on three groups of plants for 5 weeks. We want to check if there is a difference in the mean growth of each group. Using the data given below apply a one way ANOVA test at 0.05 significant level.

Fertilizer 1	**Fertilizer 2**	**Fertilizer 3**
6	8	13
8	12	9
4	9	11
5	11	8
3	6	7
4	8	12

- **Solution:**
- H0: $\mu 1 = \mu 2 = \mu 3$
- H1: The means are not equal

Fertilizer 1	**Fertilizer 2**	**Fertilizer 3**
6	8	13
8	12	9
4	9	11
5	11	8
3	6	7
4	8	12
$\overline{X1} = 5$	$\overline{X1} = 9$	$\overline{X1} = 10$

- Total mean, $\overline{X} = 8$
- $n1 = n2 = n3 = 6, k = 3$
- $SSB = 6(5 - 8)^2 + 6(9 - 8)^2 + 6(10 - 8)^2$
- $= 84$
- $df_1 = k - 1 = 2$

Fertilizer 1	**$(X - 5)^2$**	**Fertilizer 2**	**$(X - 9)^2$**	**Fertilizer 3**	**$(X - 10)^2$**
6	1	8	1	13	9
8	9	12	9	9	1
4	1	9	0	11	1
5	0	11	4	8	4
3	4	6	9	7	9
4	1	8	1	12	4
$\overline{X1} = 5$	Total = 16	$\overline{X1} = 9$	Total = 24	$\overline{X1} = 10$	Total = 28

- $SSE = 16 + 24 + 28 = 68$
- $N = 18$

- $df_2 = N - k = 18 - 3 = 15$
- $MSB = SSB / df_1 = 84 / 2 = 42$
- $MSE = SSE / df_2 = 68 / 15 = 4.53$
- ANOVA test statistic, $f = MSB / MSE = 42 / 4.53 = 9.33$
- Using the f table at $\alpha\alpha = 0.05$ the critical value is given as $F(0.05, 2, 15) = 3.68$
- As $f > F$, thus, the null hypothesis is rejected and it can be concluded that there is a difference in the mean growth of the plants.

  Answer: Reject the null hypothesis

Example 2: A trial was run to check the effects of different diets. Positive numbers indicate weight loss and negative numbers indicate weight gain. Check if there is an average difference in the weight of people following different diets using an ANOVA Table.

Low Fat	Low Calorie	Low Protein	Low Carbohydrate
8	2	3	2
9	4	5	2
6	3	4	-1
7	5	2	0
3	1	3	3

- **Solution:**
- H0: $\mu1 = \mu2 = \mu3 = \mu4$
- H1: The means are not equal

Low Fat	$(X - 6.6)^2$	Low Calorie	$(X - 3)^2$	Low Protein	$(X - 3.4)^2$	Low Carbohydrate	$(X - 1.2)^2$
8	2	2	1	3	0.2	2	0.6
9	5.8	4	1	5	2.6	2	0.6
6	0.4	3	0	4	0.4	-1	4.8
7	0.2	5	4	2	2	0	1.4
3	13	1	4	3	0.2	3	3.2

Low Fat	$(X - 6.6)^2$	Low Calorie	$(X - 3)^2$	Low Protein	$(X - 3.4)^2$	Low Carbohydrate	$(X - 1.2)^2$
$\overline{X}_1 = 6.6$	Total = 21.4	$\overline{X}_2 = 3$	Total = 10	$\overline{X}_3 = 3.4$	Total = 5.4	$\overline{X}_4 = 1.2$	Total = 10.6

- Total mean, $\overline{X} = 3.6$
- $n1 = n2 = n3 = n4 = 5$, $k = 4$
- SSB $= n1(\overline{X}_1 - \overline{X})^2 + n2(\overline{X}_2 - \overline{X})^2 + \& \; n3(\overline{X}_3 - \overline{X})^2 + n4(\overline{X}_4 - \overline{X})^2$
- $= 75.8$
- SSE = 21.4 + 10 + 5.4 + 10.6 = 47.4
- The ANOVA Table can be constructed as follows:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Value
Between Groups	$SSB = \Sigma n_j(\overline{X}_j - 3.6)2\bar{X}_j - 3.6)2 = 75.8$	$df_1 = k - 1 = 4 - 1 = 3$	$MSB = SSB / (k - 1) = 25.3$	$f = MSB / MSE = 8.43$
Error	$SSE = \Sigma\Sigma(X - \overline{X}_j)2X - \bar{X}_j)2 = 47.4$	$df_2 = N - k = 20 - 4 = 16$	$MSE = SSE / (N - k) = 3$	
Total	SST = SSB + SSE = 123.2	$df_3 = N - 1 = 19$		

- As no significance level is specified, $\alpha\alpha = 0.05$ is chosen.
- $F(0.05, 3, 16) = 3.24$
- As 8.43 > 3.24, thus, the null hypothesis is rejected and it can be concluded that there is a mean weight loss in the diets.

Answer: Reject the null hypothesis

Example 3: Determine if there is a difference in the mean daily calcium intake for people with normal bone density, osteopenia, and osteoporosis at a 0.05 alpha level. The data was recorded as follows:

Normal Density	Osteopenia	Osteoporosis

Normal Density	Osteopenia	Osteoporosis
1200	1000	890
1000	1100	650
980	700	1100
900	800	900
750	500	400
800	700	350

- **Solution:**
- Using the ANOVA test the hypothesis is set up as follows:
- H0: $\mu 1 = \mu 2 = \mu 3$
- H1: The means are not equal

Normal Density	$(X - 938.3)^2$	Osteopenia	$(X - 800)^2$	Osteoporosis	$(X - 715)^2$
1200	68,486.9	1000	40,000	890	30,625
1000	3,806.9	1100	90,000	650	4,225
980	1,738.9	700	10,000	1100	148,225
900	1,466.9	800	0	900	34,225
750	35,456.9	500	90,000	400	99,225
800	19,126.9	700	10,000	350	133,225
$\overline{X}1 =$ 938.3	Total = 130,083.3	$\overline{X}2 =$ 800	Total = 240,000	$\overline{X}3 =$ 715	Total = 449,750

- Total mean, $\overline{X} = 817.8$
- $n1 = n2 = n3 = 6, k = 3$
- $SSB = n1(\overline{X}1 - \overline{X})^2 + n2(\overline{X}2 - \overline{X})^2 + n3(\overline{X}3 - \overline{X})^2$
- $= 152,477.7$
- $SSE = 130,083.3 + 240,000 + 449,750 = 819,833.3$
- The ANOVA Table can be constructed as follows:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Value
Between Groups	SSB = $\Sigma n_j(\overline{\overline{X}}j - 817.8)^2$ = 152,477.7	df_1 = k - 1 = 3 - 1 = 2	MSB = SSB / (k - 1) = 76,238.6	f = MSB / MSE = 1.395
Error	SSE = $\Sigma\Sigma(X - \overline{\overline{X}}j)^2$ = 819,833.3	df_2 = N - k = 18 - 3 = 15	MSE = SSE / (N - k) = 54,655.5	
Total	SST = SSB + SSE = 972,311	df_3 = N - 1 = 17		

- Using the F table the critical value is F(0.05, 2, 15) = 3.68
- As 1.395 < 3.68, the null hypothesis cannot be rejected and it is concluded that there is not enough evidence to prove that the mean daily calcium intake of the three groups is different.
- **Answer:** Do not reject the null hypothesis

Correlation and Covariance:-Pearson Correlation Coefficient, Spearman's Rank Correlation,

What is Covariance?

Covariance is a statistical measure that **indicates the direction of the linear relationship between two variables**. It assesses how much two variables change together from their mean values.

Types of Covariance:

- **Positive Covariance**: When one variable increases, the other variable tends to increase as well, and vice versa.
- **Negative Covariance**: When one variable increases, the other variable tends to decrease.
- **Zero Covariance**: There is no linear relationship between the two variables; they move independently of each other.

Covariance is calculated by taking the average of the product of the deviations of each variable from their respective means. It is useful for understanding the direction of the relationship but not its strength, as its magnitude depends on the units of the variables.

It is an essential tool for understanding how variables change together and is widely used in various fields, including finance, economics, and science.

Covariance:

1. It is the relationship between a pair of random variables where a change in one variable causes a change in another variable.
2. It can take any value between – infinity to +infinity, where the negative value represents the negative relationship whereas a positive value represents the positive relationship.
3. It is used for the linear relationship between variables.
4. It gives the direction of relationship between variables.

Covariance Formula

For Population:

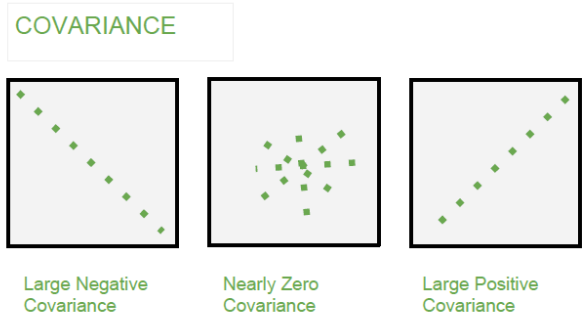
$$Covri(x,y) = \frac{\sum_{i=1}^{n}(x_i - x')\ (y_i - y')}{n}$$

For Sample:

$$Covari(x,y) = \frac{\sum_{i=1}^{n}(x_i - x')\ (y_i - y')}{n-1}$$

Here, x' and y' = mean of given sample set n = total no of sample xi and yi = individual sample of set

Example –



What is Correlation?

Correlation is a **standardized measure of the strength and direction of the linear relationship between two variables**. It is derived from covariance and **ranges between -1 and 1**. Unlike covariance, which only indicates the direction of the relationship, correlation provides a standardized measure.

- **Positive Correlation (close to +1)**: As one variable increases, the other variable also tends to increase.
- **Negative Correlation (close to -1)**: As one variable increases, the other variable tends to decrease.
- **Zero Correlation**: There is no linear relationship between the variables.

The correlation coefficient ρ\rhoρ (rho) for variables X and Y is defined as:

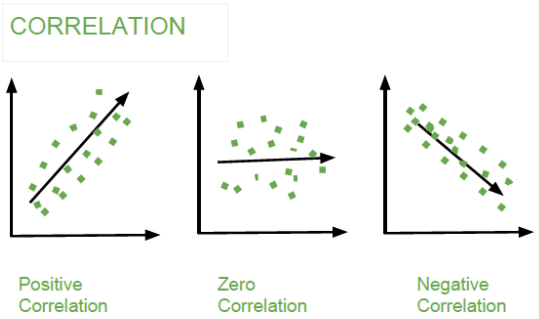
1. It show whether and how strongly pairs of variables are related to each other.
2. Correlation takes values between -1 to +1, wherein values close to +1 represents strong positive correlation and values close to -1 represents strong negative correlation.
3. In this variable are indirectly related to each other.
4. It gives the direction and strength of relationship between variables.

Correlation Formula

$$Corr(x, y) = \frac{\sum_{i=1}^{n}(x_i - x')\,(y_i - y')}{\sqrt{\sum_{i=1}^{n}(x_i - x')^2 \sum_{i=1}^{n}(y_i - y')^2}}$$

Here, x' and y' = mean of given sample set n = total no of sample xi and yi = individual sample of set

Example –



Difference between Covariance and Correlation

Covariance	Correlation
Covariance is a measure of how much two random variables vary together	Correlation is a statistical measure that indicates how strongly two variables are related.
Involves the relationship between two variables or data sets	Involves the relationship between multiple variables as well

Covariance	Correlation
Lie between -infinity and +infinity	Lie between -1 and +1
Measure of correlation	Scaled version of covariance
Provides direction of relationship	Provides direction and strength of relationship
Dependent on scale of variable	Independent on scale of variable
Have dimensions	Dimensionless

Applications of Covariance

- **Portfolio Management in Finance**: Covariance is used to measure how different stocks or financial assets move together, aiding in portfolio diversification to minimize risk.
- **Genetics**: In genetics, covariance can help understand the relationship between different genetic traits and how they vary together.
- **Econometrics**: Covariance is employed to study the relationship between different economic indicators, such as the relationship between GDP growth and inflation rates.
- **Signal Processing**: Covariance is used to analyze and filter signals in various forms, including audio and image signals.
- **Environmental Science**: Covariance is applied to study relationships between environmental variables, such as temperature and humidity changes over time

Applications of Correlation

- **Market Research**: Correlation is used to identify relationships between consumer behavior and sales trends, helping businesses make informed marketing decisions.
- **Medical Research**: Correlation helps in understanding the relationship between different health indicators, such as the correlation between blood pressure and cholesterol levels.
- **Weather Forecasting**: Correlation is used to analyze the relationship between various meteorological variables, such as temperature and humidity, to improve weather predictions.
- **Machine Learning**: Correlation analysis is used in feature selection to identify which variables have strong relationships with the target variable, improving model accuracy.

Difference between Spearman and Pearson Correlation

Spearman	Pearson
Spearman correlation is used for ordinal, interval, or ratio data.	Pearson correlation is used for continuous, numerical data.

Spearman	Pearson
Non-parametric and does not assume a specific distribution.	Assumes that the data follows a bivariate normal distribution.
Detects monotonic relationships, including non-linear associations.	Assumes linear relationships.
Less sensitive to outliers.	Sensitive to outliers.

Pearson correlation coefficient:

The **Pearson correlation coefficient (r)** is the most common way of measuring a linear correlation. It is a number between –1 and 1 that measures the strength and direction of the relationship between two variables.

Pearson correlation coefficient (r)	**Correlation type**	**Interpretation**	**Example**
Between 0 and 1	Positive correlation	When one variable changes, the other variable changes in the **same direction**.	Baby length & weight: The longer the baby, the heavier their weight.
0	No correlation	There is **no relationship** between the variables.	Car price & width of windshield wipers: The price of a car is not related to the width of its windshield wipers.
Between 0 and –1	Negative correlation	When one variable changes, the other variable changes in the **opposite direction**.	Elevation & air pressure: The higher the elevation, the lower the air pressure.

Although interpretations of the relationship strength (also known as effect size) vary between disciplines, the table below gives general rules of thumb:

Pearson correlation coefficient (r) value	**Strength**	**Direction**
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive

Pearson correlation coefficient (r) value	Strength	Direction
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and –.3	Weak	Negative
Between –.3 and –.5	Moderate	Negative
Less than –.5	Strong	Negative

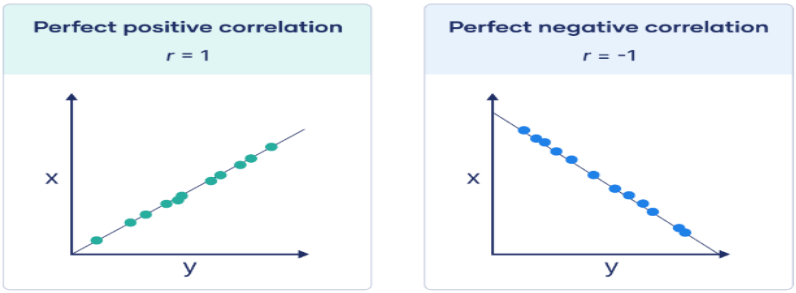
The Pearson correlation coefficient is also an inferential statistic, meaning that it can be used to test statistical hypotheses. Specifically, we can test whether there is a significant relationship between two variables.

Visualizing the Pearson correlation coefficient

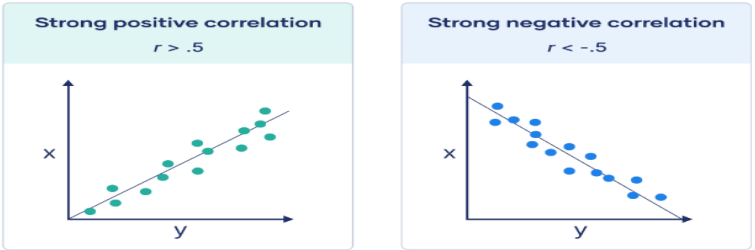
Another way to think of the Pearson correlation coefficient (r) is as a measure of how close the observations are to a line of best fit.

The Pearson correlation coefficient also tells you whether the slope of the line of best fit is negative or positive. When the slope is negative, r is negative. When the slope is positive, r is positive.

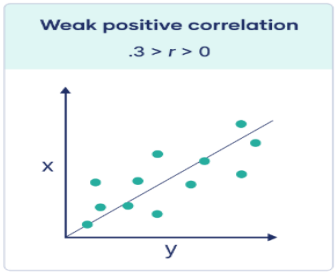
When r is 1 or –1, all the points fall exactly on the line of best fit:

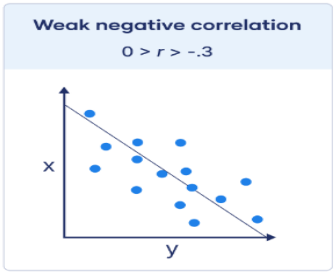


When r is greater than .5 or less than –.5, the points are close to the line of best fit:

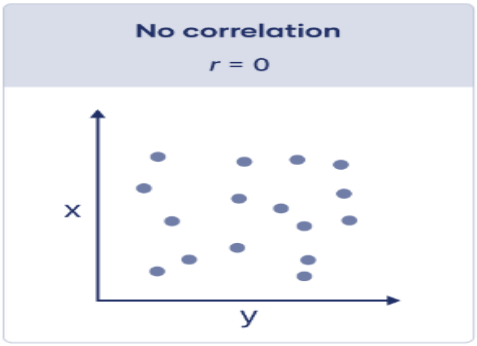


When r is between 0 and .3 or between 0 and –.3, the points are far from the line of best fit:





When r is 0, a line of best fit is not helpful in describing the relationship between the variables:



When to use the Pearson correlation coefficient

The Pearson correlation coefficient (r) is one of several correlation coefficients that you need to choose between when you want to measure a correlation. The Pearson correlation coefficient is a good choice when **all** of the following are true:

- **Both variables are quantitative:** You will need to use a different method if either of the variables is qualitative.
- **The variables are normally distributed:** You can create a histogram of each variable to verify whether the distributions are approximately normal. It's not a problem if the variables are a little non-normal.
- **The data have no outliers:** Outliers are observations that don't follow the same patterns as the rest of the data. A scatterplot is one way to check for outliers—look for points that are far away from the others.
- **The relationship is linear:** "Linear" means that the relationship between the two variables can be described reasonably well by a straight line. You can use a scatterplot to check whether the relationship between two variables is linear.

Pearson vs. Spearman's rank correlation coefficients

Spearman's rank correlation coefficient is another widely used correlation coefficient. It's a better choice than the Pearson correlation coefficient when **one or more** of the following is true:

- The variables are ordinal.
- The variables aren't normally distributed.
- The data includes outliers.
- The relationship between the variables is non-linear **and** monotonic.

Calculating the Pearson correlation coefficient

Below is a formula for calculating the Pearson correlation coefficient (r):

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Example: Dataset

1. Imagine that you're studying the relationship between newborns' weight and length. You have the weights and lengths of the 10 babies born last month at your local hospital. After you convert the imperial measurements to metric, you enter the data in a table:

Weight (kg)	Length (cm)
3.63	53.1
3.02	49.7
3.82	48.4
3.42	54.2
3.59	54.9
2.87	43.7
3.03	47.2
3.46	45.2
3.36	54.4
3.3	50.4

Step 1: Calculate the sums of x and y

Start by renaming the variables to "x" and "y." It doesn't matter which variable is called x and which is called y—the formula will give the same answer either way.

Next, add up the values of x and y. (In the formula, this step is indicated by the Σ symbol, which means "take the sum of".)

Example: Calculating the sums of x and y

Weight = x

Length = y

$\Sigma x = 3.63 + 3.02 + 3.82 + 3.42 + 3.59 + 2.87 + 3.03 + 3.46 + 3.36 + 3.30$

$\Sigma x = 33.5$

$\Sigma y = 53.1 + 49.7 + 48.4 + 54.2 + 54.9 + 43.7 + 47.2 + 45.2 + 54.4 + 50.4$

$\Sigma y = 501.2$

Step 2: Calculate x^2 and y^2 and their sums

Create two new columns that contain the squares of x and y. Take the sums of the new columns.

Example: Calculating x^2 and y^2 and their sums

x	**y**	**x^2**	**y^2**
3.63	53.1	(3.63)2 = 13.18	(53.1)2 = 2 819.6
3.02	49.7	9.12	2 470.1
3.82	48.4	14.59	2 342.6
3.42	54.2	11.7	2 937.6
3.59	54.9	12.89	3 014
2.87	43.7	8.24	1 909.7
3.03	47.2	9.18	2 227.8
3.46	45.2	11.97	2 043
3.36	54.4	11.29	2 959.4
3.3	50.4	10.89	2 540.2

$\Sigma x^2 = 13.18 + 9.12 + 14.59 + 11.70 + 12.89 + 8.24 + 9.18 + 11.97 + 11.29 + 10.89$

$\Sigma x^2 = 113.05$

$\Sigma y^2 = 2\,819.6 + 2\,470.1 + 2\,342.6 + 2\,937.6 + 3\,014.0 + 1\,909.7 + 2\,227.8 + 2\,043.0 + 2\,959.4 + 2\,540.2$

$\Sigma y^2 = 25\,264$

Step 3: Calculate the cross product and its sum

In a final column, multiply together x and y (this is called the cross product). Take the sum of the new column.

Example: Calculating the cross product and its sum

x	y	x^2	y^2	xy (x*y)
3.63	53.1	13.18	2 819.6	3.63 * 53.1 = 192.8
3.02	49.7	9.12	2 470.1	150.1
3.82	48.4	14.59	2 342.6	184.9
3.42	54.2	11.7	2 937.6	185.4
3.59	54.9	12.89	3 014	197.1
2.87	43.7	8.24	1 909.7	125.4
3.03	47.2	9.18	2 227.8	143
3.46	45.2	11.97	2 043	156.4
3.36	54.4	11.29	2 959.4	182.8
3.3	50.4	10.89	2 540.2	166.3

Σxy = 192.8 + 150.1 + 184.9 + 185.4 + 197.1 + 125.4 + 143.0 + 156.4 + 182.8 + 166.3

Σxy = 1 684.2

Step 4: Calculate r

Use the formula and the numbers you calculated in the previous steps to find r.

Example: Calculating r$n = 10$

$$\sum x = 33.5$$

$$\sum y = 501.2$$

$$\sum x^2 = 113.05$$

$$\sum y^2 = 25\,264$$

$$\sum xy = 1\,684.2$$

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

$$r = \frac{10\sum 1\,684.2 - (33.5)(501.2)}{\sqrt{[(10)(113.05) - (33.5)^2][(10)(25\,264) - (501.2)^2]}}$$

$$r = \frac{16\,842 - 16\,790.2)}{\sqrt{[1\,130.5 - 1\,122.25][252\,640 - 251\,201.4]}}$$

$$r = \frac{51.8}{\sqrt{11\,868.45}}$$

$$r = 0.47$$

Example 1: There is some correlation coefficient that was given to tell whether the variables are positive or negative?
0.69, 0.42, -0.23, -0.99
Solution:
The given correlation coefficient is as follows:
0.69, 0.42, -0.23, -0.99
Tell whether the relationship is negative or positive
0.69: The relationship between the variables is a strong positive relationship
0.42: The relationship between the variables is a strong positive relationship
-0.23: The relationship between the variables is a weak negative relationship
-0.99: The relationship between the variables is a very strong negative relationship

Example 2: Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:
X = 10, 13, 15 ,17 ,19
and
Y = 5,10,15,20,25.
Solution:
Given variables are,
X = 10, 13, 15 ,17 ,19
and
Y = 5,10,15,20,25.
To, find the correlation coefficient of the following variables Firstly a table is to be constructed as follows, to get the values required in the formula also add all the values in the columns to get the values used in the formula.

X	Y	XY	X^2	Y^2
10	5	50	100	25
13	10	130	169	100
15	15	225	225	225
17	20	340	289	400

X	Y	XY	X^2	Y^2
19	25	475	362	625
∑74	∑75	∑1103	∑1144	∑1375

$\sum xy = 1103$
$\sum x = 74$
$\sum y = 75$
$\sum x^2 = 1144$
$\sum y^2 = 1375$
n = 5
Put all the values in the Pearson's correlation coefficient formula:-
$R = n(\sum xy) – (\sum x)(\sum y) / \sqrt{} [n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2$
$R = 5(1103) – (74)(75) / \sqrt{} [5(1144)-(74)^2][5(1375)-(75)^2]$
$R = -35 / \sqrt{}[244][1250]$
R = -35/552.26
R = 0.0633
The correlation coefficient is 0.064

Example 3: Calculate the correlation coefficient for the following table with the help of Pearson's correlation coefficient formula:

SUBJECT	**AGE X**	**Weight Y**
1	**40**	**99**
2	**25**	**79**
3	**22**	**69**
4	**54**	**89**

Solution:
Make a table from the given data and add three more columns of XY, X^2, and Y^2. also add all the values in the columns to get $\sum xy$, $\sum x$, $\sum y$, $\sum x^2$, and $\sum y^2$ and n =4.

SUBJECT	AGE X	Weight Y	XY	X^2	Y^2
1	40	99	3960	1600	9801
2	25	79	1975	625	6241
3	22	69	1518	484	4761
4	54	89	4806	2916	7921

SUBJECT	AGE X	Weight Y	XY	X^2	Y^2
∑	151	336	12259	5625	28724

$\sum xy = 12258$
$\sum x = 151$
$\sum y = 336$
$\sum x^2 = 5625$
$\sum y^2 = 28724$
$n = 4$
Put all the values in the Pearson's correlation coefficient formula:-
$R = n(\sum xy) - (\sum x)(\sum y) / \sqrt{[n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2}$
$R = 4(12258) - (151)(336) / \sqrt{[4(5625)-(151)^2][4(28724)-(336)^2]}$
$R = -1704 / \sqrt{[-301][-2000]}$
$R = -1704/775.886$
$R = -2.1961$
The correlation coefficient is -2.196

Example 4: Calculate the correlation coefficient for the following data with the help of Pearson's correlation coefficient formula:
X = 5 ,9 ,14, 16
and
Y = 6, 10, 16, 20 .
Solution:
Given variables are,
X = 5 ,9 ,14, 16
and
Y = 6, 10, 16, 20 .
To, find the correlation coefficient of the following variables Firstly a table to be constructed as follows, to get the values required in the formula
also, add all the values in the columns to get the values used in the formula.

X	Y	XY	X^2	Y^2
5	6	30	25	36
9	10	90	81	100
14	16	224	196	256
16	20	320	256	400
∑ 44	∑ 52	∑ 664	∑ 558	∑ 792

$\sum xy = 664$
$\sum x = 44$
$\sum y = 52$
$\sum x^2 = 558$
$\sum y^2 = 792$
$n = 4$

Put all the values in the Pearson's correlation coefficient formula:-
$R= n(\sum xy) – (\sum x)(\sum y) / \sqrt{} [n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2$
$R= 4(664) – (44)(52) / \sqrt{} [4(558)-(44)^2][4(792)-(52)^2]$
$R= 368 / \sqrt{}[296][464]$
R=368/370.599
R=0.994
The correlation coefficient is 0.994

Example 5: Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:
X = 21,31,25,40,47,38
and
Y = 70,55,60,78,66,80
Solution:
Given variables are,
X = 21,31,25,40,47,38
and
Y = 70,55,60,78,66,80
To, find the correlation coefficient of the following variables Firstly a table is to be constructed as follows, to get the values required in the formula also add all the values in the columns to get the values used in the formula.

X	Y	XY	X^2	Y^2
21	70	1470	441	4900
31	55	1705	961	3025
25	60	1500	625	3600
40	78	3120	1600	6084
47	66	3102	2209	4356
38	80	3040	1444	6400
∑202	∑409	∑13937	∑7280	∑28265

$\sum xy= 13937$
$\sum x=202$
$\sum y=409$
$\sum x^2 =7280$
$\sum y^2 =28265$
n =6
Put all the values in the Pearson's correlation coefficient formula:-
$R= n(\sum xy) – (\sum x)(\sum y) / \sqrt{} [n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2$
$R= 6(13937) – (202)(409) / \sqrt{} [6(7280)-(202)^2][6(28265)-(409)^2]$
$R= 1004 / \sqrt{}[2876][2909]$
R=1004 / 2892.452938
R=-0.3471

The correlation coefficient is -0.3471

Example 6: Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:

SUBJECT	**Height X**	**Weight Y**
1	**43**	**78**
2	**24**	**68**
3	**26**	**85**
4	**35**	**67**

Solution:
Make a table from the given data and add three more columns of XY , X^2 and Y^2 and add all the values in the columns to get $\sum xy$, $\sum x$, $\sum y$, $\sum x^2$ and $\sum y^2$ and n =4.

SUBJECT	Height X	Weight Y	XY	X^2	Y^2
1	43	78	3354	1849	6084
2	24	68	1632	576	4624
3	26	85	2210	676	7225
4	35	67	2345	1225	4489
$\sum$	128	298	9541	4317	22422

$\sum xy$= 9541
$\sum x$=128
$\sum y$=298
$\sum x^2$ =4317
$\sum y^2$ 22422
n =4
Put all the values in the Pearson's correlation coefficient formula:-
$R= n(\sum xy) – (\sum x)(\sum y) / \sqrt{} [n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2$
$R= 4(9541) – (128)(298) / \sqrt{} [4(4317)-(128)^2][4(22422)-(298)^2]$
$R= 20 / \sqrt{} [884][884]$
R=20/884
R=0.02262
The correlation coefficient is 0.02262

Spearman's Rank

Spearman's Rank Correlation Coefficient or Spearman's Rank Difference Method or Formula is a method of calculating the correlation coefficient of qualitative variables and was developed in **1904 by Charles Edward Spearman**.

In other words, the formula determines the correlation coefficient of variables like beauty, ability, honesty, etc., whose quantitative measurement is not possible. Therefore, these attributes are ranked or put in the order of their preference.

Or

Spearman's rank correlation measures the strength and direction of association between two ranked variables. It basically gives the measure of monotonicity of the relation between two variables i.e. how well the relationship between two variables could be represented using a monotonic function.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

In the given formula,
P = Coefficient of rank correlation
d = Rank differences
n = Number of variables

The Spearman Rank Correlation can take a value from +1 to -1 where,

- A value of +1 means a perfect association of rank
- A value of 0 means that there is no association between ranks
- A value of -1 means a perfect negative association of rank

Example of Spearman's Rank Correlation

Consider the score of 5 students in Maths and Science that are mentioned in the table.

Students	Maths	Science
A	35	24
B	20	35
C	49	39
D	44	48
E	30	45

Step 1: Create a table for the given data.

Step 2: Rank both the data in descending order. The highest marks will get a rank of 1 and the lowest marks will get a rank of 5.

Step 3: Calculate the difference between the ranks (d) and the square value of d.

Step 4: Add all your d square values.

Students	Maths	Rank	Science	Rank	d	d square
A	35	3	24	5	2	4
B	20	5	35	4	1	1
C	49	1	39	3	2	4
D	44	2	48	1	1	1
E	30	4	45	2	2	4
						14

Step 5: Insert these values into the formula.

$$\rho = 1 - \frac{6\Sigma\, d_i^2}{n(n^2 - 1)}$$

= 1 - (6 * 14) / 5(25 - 1)

= 0.3

The Spearman's Rank Correlation for the given data is 0.3. The value is near 0, which means that there is a weak correlation between the two ranks.

2. To calculate a Spearman rank-order correlation on data without any ties we will use the following data:

	Marks									
English	56	75	45	71	62	64	58	80	76	61
Maths	66	70	40	60	65	56	59	77	67	63

We then complete the following table:

English (mark)	Maths (mark)	Rank (English)	Rank (maths)	d	d^2
56	66	9	4	5	25
75	70	3	2	1	1

English (mark)	Maths (mark)	Rank (English)	Rank (maths)	d	d^2
45	40	10	10	0	0
71	60	4	7	3	9
62	65	6	5	1	1
64	56	5	9	4	16
58	59	8	8	0	0
80	77	1	1	0	0
76	67	2	3	1	1
61	63	7	6	1	1

Where d = difference between ranks and d^2 = difference squared.

We then calculate the following:

$$\sum d_i^2 = 25 + 1 + 9 + 1 + 16 + 1 + 1 = 54$$

We then substitute this into the main equation with the other information as follows:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$\rho = 1 - \frac{6 \times 54}{10(10^2 - 1)}$$

$$\rho = 1 - \frac{324}{990}$$

$$\rho = 1 - 0.33$$

$$\rho = 0.67$$

Problem 1:

In a contest for search for Ms. Puerto Princesa City 2020, two judges gave their ratings to 8 candidates. Transform the ratings to ranks and compute the coefficient of rank correlation. Interpret the results.

Candidate	Judge 1	Judge 2
1	97	93
2	96	96
3	94	97
4	89	94
5	88	91
6	87	89
7	84	88
8	84	84

Solution:

Candidate	Judge 1 (x)	Judge 2 (y)	Ranks for Judge 1 (x)	Ranks for Judge 2 (y)	$d_1 = x_1 - y_1$	d_{12}
1	97	93	1	4	-3	9
2	96	96	2	2	0	0
3	94	97	3	1	2	4
4	89	94	4	3	1	1
5	88	91	5	5	0	0
6	87	89	6	6	0	0
7	84	88	7.5	7	0.5	0.25
8	84	84	7.5	8	-0.5	0.25
						$\Sigma d^2 = 14.5$

$$\rho = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$p = 1 - \frac{6(14.5)}{8(8^2-1)}$$

$$p = 1 - \frac{87}{8(64-1)}$$

$$p = 1 - \frac{87}{8(63)}$$

$$p = 1 - \frac{87}{504} \qquad p = 1 - 0.17 \qquad \boxed{p = 0.83}$$

Problem 2:

The Land Transformation Office (LTO) believes that motorcycle riders who do not wear helmet while driving are at high risk when an accident happens. They gathered the following data in different regions per month.

Region	Number of accidents (riders with helmet)	Number of accidents (riders without helmet)
1	20	21
2	12	20
3	16	17
4	15	20
5	17	18
6	28	30
7	23	30
8	12	15
9	15	15
10	13	14

Solution:

Regions	With Helmet (x)	Without Helmet (y)	Ranks (With Helmet) (x)	Ranks for (Without Helmet) (y)	$d_1 = x_1 - y_1$	d_{12}
1	20	21	3	3	0	0
2	12	20	9.5	4.5	5	25
3	16	17	5	7	-2	4
4	15	20	6.5	4.5	2	4
5	17	18	4	6	-2	4
6	28	30	1	1.5	-0.5	0.25
7	23	30	2	1.5	0.5	0.25
8	12	15	9.5	8.5	1	1
9	15	15	6.5	8.5	-2	4
10	13	14	8	10	-2	4

$\Sigma d^2 = 46.5$

Solution:

$$p = 1 - \frac{6(46.5)}{10(10^2 - 1)}$$

$$p = 1 - \frac{279}{10(100 - 1)}$$

$$p = 1 - \frac{279}{10(99)}$$

$$p = 1 - \frac{279}{990} \qquad p = 1 - 0.28 \qquad p = 0.72$$

Problem 3:

The hypothesis tested that prices should decrease with the distance from the key area of the city surrounding the Puerto Princesa City Coliseum. They gathered the sampling price of 1 litter bottled water at every convenience store.

Convenience Store	Distance from Coliseum (m)	Rank in Distance	Price of IL Bottled Water (Peso)	Rank in Price	Difference Between Rank (d)	d_{12}
1	40	8	35	2	6	36
2	57	7	30	5.5	1.5	2.25
3	89	6	50	1	5	25
4	120	5	29	7	-2	4
5	147	4	30	5.5	-1.5	2.25
6	210	3	31	3.5	-0.5	0.25
7	229	2	28	8	-6	36
8	506	1	31	3.5	-2.5	6.25
						$\Sigma d^2 = 112$

Solution:

$$p = 1 - \frac{6(112)}{8(8^2-1)} \qquad p = 1 - \frac{672}{8(63)}$$

$$p = 1 - \frac{672}{8(64-1)} \qquad p = 1 - \frac{672}{504} \qquad p = 1 - 1.33 \qquad p = -0.33$$

Problem 4:

Determine whether the following Cueshe' album popularity in terms of sales were related to when they released. Determine whether release is related to popularity.

Album	Year	Release Rank	Sales Rank	d	d^2
Half Empty, Half Full	2005	1	7	-6	36
Back to me	2006	2	6	-4	16
Life	2007	3	8	-5	25
Superstar	2008	4	1	3	9
Ala-Ala	2009	5	4	1	1
Drift	2010	6	3	3	9
BDM	2011	7	2	5	25
Over	2012	8	5	3	9
					$\Sigma d^2 = 130$

Solution:

$$p = 1 - \frac{6(130)}{8(8^2-1)}$$

$$p = 1 - \frac{780}{8(64-1)}$$

$$p = 1 - \frac{780}{8(63)}$$

$$p = 1 - \frac{780}{504} \qquad p = 1 - 1.55 \qquad p = -0.55$$

Covariance and Its Interpretation

What Is Covariance?

Covariance is a statistical term that refers to a systematic relationship between two random variables in which a change in the other reflects a change in one variable.

The covariance value can range from -∞ to +∞, with a negative value indicating a negative relationship and a positive value indicating a positive relationship.

The greater this number, the more reliant the relationship. Positive covariance denotes a direct relationship and is represented by a positive number.

A negative number, on the other hand, denotes negative covariance, which indicates an inverse relationship between the two variables. Covariance is great for defining the type of relationship, but it's terrible for interpreting the magnitude.

Let Σ(X) and Σ(Y) be the expected values of the variables, the covariance formula can be represented as:

$$\text{Covariance}(x, y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Where,

- xi = data value of x
- yi = data value of y
- x̄ = mean of x
- ȳ = mean of y
- N = number of data values.

Formula:

Population Covariance Formula

$$\text{Cov}(x,y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

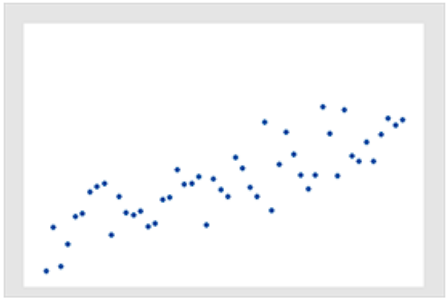
$$\text{Cov}(x,y) = \frac{\sum(x_i - \bar{x})(y_i - y)}{N - 1}$$

Types of Covariance

Covariance can be either positive or negative. It is classified into two sorts based on this:

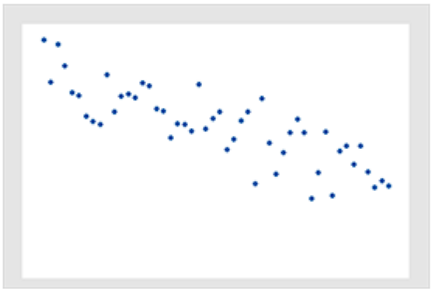
Positive Covariance

A positive covariance between two variables indicates that they are heading in the same direction. The variables, in this case, behave similarly. That is, if the values of one variable (more or smaller) correspond to the values of another variable, they are said to be in positive covariance.



Negative Covariance

When two variables have a negative covariance, the variables shift in the opposite direction. It is the inverse of positive covariance, in which higher values of one variable correlate to lower values of another and vice versa.



Application Of Covariance

- Simulating systems with multiple correlated variables is done using Cholesky decomposition. A covariance matrix helps determine the Cholesky decomposition because it is positive semi-definite. The matrix is decomposed by the product of the lower matrix and its transpose.
- To reduce the dimensions of large data sets, principal component analysis is used. To perform principal component analysis, an eigen decomposition is applied to the covariance matrix

What Is A Covariance Matrix?

A covariance matrix is a square matrix that illustrates the variance of dataset elements and the covariance between two datasets. Variance is a measure of dispersion defined as data spread from the provided dataset's mean. Covariance between two variables is calculated and used to measure how the two variables fluctuate together.

Covariance vs. Correlation

Basis for comparison	Covariance	Correlation
Definition	Covariance is an indicator of the extent to which 2 random variables are dependent on each other. A higher number denotes higher dependency.	Correlation is a statistical measure that indicates how strongly two variables are related.
Values	The value of covariance lies in the range of -∞ and +∞.	Correlation is limited to values between the range -1 and +1
Change in scale	Affects covariance	Does not affect the correlation
Unit-free measure	No	Yes

Or

Difference between Covariance and Correlation

Covariance	Correlation
Covariance is a measure of how much two random variables vary together	Correlation is a statistical measure that indicates how strongly two variables are related.

Covariance	Correlation
Involves the relationship between two variables or data sets	Involves the relationship between multiple variables as well
Lie between -infinity and +infinity	Lie between -1 and +1
Measure of correlation	Scaled version of covariance
Provides direction of relationship	Provides direction and strength of relationship
Dependent on scale of variable	Independent on scale of variable
Have dimensions	Dimensionless

Covariance vs. Correlation Formula:

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Where:

- **ρ(X,Y)** – the correlation between the variables X and Y
- **Cov(X,Y)** – the covariance between the variables X and Y
- **σ_X** – the standard deviation of the X-variable
- **σ_Y** – the standard deviation of the Y-variable

Covariance Example

Question:

Calculate the coefficient of covariance for the following data:

X	2	8	18	20	28	30

Y	5	12	18	23	45	50

Solution:

Number of observations = 6

Mean of X = 17.67

Mean of Y = 25.5

Cov(X, Y) = (⅙) [(2 – 17.67)(5 – 25.5) + (8 – 17.67)(12 – 25.5) + (18 – 17.67)(18 – 25.5) + (20 – 17.67)(23 – 25.5) + (28 – 17.67)(45 – 25.5) + (30 – 17.67)(50 – 25.5)]

= 157.83

Example of Covariance

John is an investor. His portfolio primarily tracks the performance of the S&P 500 and John wants to add the stock of ABC Corp. Before adding the stock to his portfolio, he wants to assess the directional relationship between the stock and the S&P 500.

John does not want to increase the unsystematic risk of his portfolio. Thus, he is not interested in owning securities in the portfolio that tend to move in the same direction.

John can calculate the covariance between the stock of ABC Corp. and S&P 500 by following the steps below:

1. Obtain the data.

First, John obtains the figures for both ABC Corp. stock and the S&P 500. The prices obtained are summarized in the table below:

	S&P 500	ABC Corp.
2013	1,692	68
2014	1,978	102
2015	1,884	110
2016	2,151	112
2017	2,519	154

2. Calculate the mean (average) prices for each asset.

$$\text{Mean (S\&P 500)} = \frac{1{,}692 + 1{,}978 + 1{,}884 + 2{,}151 + 2{,}519}{5} = 2{,}044.80$$

$$\text{Mean (ABC Corp.)} = \frac{68 + 102 + 110 + 112 + 154}{5} = 109.20$$

3. For each security, find the difference between each value and mean price.

			Step 3		Step 4
	S&P 500	ABC Corp.	a	b	a x b
2013	1,692	68	-352.80	-41.20	14,535.36
2014	1,978	102	-66.80	-7.20	480.96
2015	1,884	110	-160.80	0.80	-128.64
2016	2,151	112	106.20	2.80	297.36
2017	2,519	154	474.20	44.80	21,244.16
Mean	**2,044.80**	**109.20**	**Sum**		**36,429.20**

4. Multiply the results obtained in the previous step.

5. Using the number calculated in step 4, find the covariance.

$$\text{Cov(S\&P 500, ABC Corp.)} = \frac{36{,}429.20}{5 - 1} = 9{,}107.30$$

In such a case, the positive covariance indicates that the price of the stock and the S&P 500 tend to move in the same direction.

Question: The table below describes the rate of economic growth (xi) and the rate of return on the S&P 500 (y_i). Using the covariance formula, determine whether economic growth and S&P 500 returns have a positive or inverse relationship. Before you compute the covariance, calculate the mean of x and y.

Economic Growth % (x_i)	S&P 500 Returns % (y_i)
2.1	8
2.5	12
4.0	14
3.6	10

x = 2.1, 2.5, 4.0, and 3.6 (economic growth)

y = 8, 12, 14, and 10 (S&P 500 returns)

Find $\bar{x}$ and $\bar{y}$.

Solution:

$\bar{x} =$

$\frac{\sum x_i}{n}$

$\bar{x} =$

$\frac{2.1+2.5+4+3.6}{4}$

$\bar{x} =$

$\frac{12.2}{4}$

$\bar{x} = 3.1$

$\bar{y} =$

$\frac{\sum x_i}{n}$

$\bar{y} =$

$\frac{8+12+14+10}{4}$

$\bar{y} =$

$\frac{44}{4}$

$\bar{y} = 11$

Now, substitute these values into the covariance formula to determine the relationship between economic growth and S&P 500 returns.

xi	yi	xi – $\bar{x}$	yi – $\bar{y}$
2.1	8	-1	-3
2.5	12	-0.6	1
4.0	14	0.9	3
3.6	10	0.5	-1

$$Cov(x,y) = \frac{(-1)(-3)+(-0.6)1+(0.9)3+(0.5)(-1)}{4-1}$$
$$= \frac{3-0.6+2.7-0.5}{3}$$
$$= \frac{4.6}{3} = 1.533$$

Example 1: Find covariance for following data set x = {2,5,6,8,9}, y = {4,3,7,5,6}

Solution:

Given data sets x = {2,5,6,8,9}, y = {4,3,7,5,6} and N = 5

Mean(x) = (2 + 5 + 6 + 8 + 9) / 5

= 30 / 5

= 6

Mean(y) = (4 + 3 +7 + 5 + 6) / 5

= 25 / 5

= 5

Sample covariance Cov(x,y) = $\sum(x_i - x) \times (y_i - y)/ (N - 1)$

= [(2 - 6)(4 - 5) + (5 - 6)(3 - 5) + (6 - 6)(7 - 5) + (8 - 6)(5 - 5) + (9 - 6)(6 - 5)] / 5 - 1

= 4 + 2 + 0 + 0 + 3 / 4

= 9 / 4

= 2.25

Population covariance Cov(x,y) = $\sum(x_i - x) \times (y_i - y)/ (N)$

= [(2 - 6)(4 - 5) + (5 - 6)(3 - 5) + (6 - 6)(7 - 5) + (8 - 6)(5 - 5) + (9 - 6)(6 - 5)] / 5

= 4 + 2 + 0 + 0 + 3 /

= 9 / 5

= 1.8

Answer: The sample covariance is 2.25 and the population covariance is 1.8.

Example 2: Using the covariance formula, find covariance for following data set x = {5,6,8,11,4,6}, y = {1,4,3,7,9,12}.

Given data sets x = {5,6,8,11,4,6}, y = {1,4,3,7,9,12} and N = 6

Mean(x) = (5 + 6 + 8 + 11 + 4 + 6) / 6

= 40 / 6

= 6.67

Mean(y) = (1 + 4 + 3 + 7 + 9 + 12) / 6

= 36 / 6

= 6

Using sample covariance formula $Cov(x,y) = \sum(x_i - x) \times (y_i - y)/ (N - 1)$

$= - 0.4$

Population covariance $Cov(x,y) = \sum(x_i - x) \times (y_i - y)/ (N)$

$= - 0.33$

Answer: The sample covariance is -0.4 and the population covariance is -0.33.

Example 3: Find covariance for following data set x = {13,15,17,18,19}, y = {10,11, 12,14,16} using the covariance formula.

Solution:

Given data sets x = {13,15,17,18,19}, y = {10,11,12,14,16} and N = 5

Mean(x) = (13 + 15 + 17 + 18 + 19) / 5

= 82 / 5

= 16.4

Mean(y) = (10 + 11 +12 + 14 + 16) / 5

= 63 / 5

= 12.6

Sample covariance $Cov(x,y) = \sum(x_i - x) \times (y_i - y)/ (N - 1)$

= [(16.4 - 13)(12.6 - 10) + (16.4 - 15)(12.6 - 11) + (16.4 - 17)(12.6 - 12) + (16.4 - 18)(12.6 - 14) + (16.4 - 19)(12.6 - 16)] / 5 - 1

= (8.84 + 2.24 - 0.36 + 2.24 + 8.84) / 4

= 21.8 / 4

= 5.45

Population covariance $Cov(x,y) = \sum(x_i - x) \times (y_i - y)/ (N)$

= [(16.4 - 13)(12.6 - 10) + (16.4 - 15)(12.6 - 11) + (16.4 - 17)(12.6 - 12) + (16.4 - 18)(12.6 - 14) + (16.4 - 19)(12.6 - 16)] / 5

= (8.84 + 2.24 - 0.36 + 2.24 + 8.84) / 5

= 21.8 / 5

= 4.36

Answer: The sample covariance is 5.45 and the population covariance is 4.36.